Data Analysis: Growth Accounting

Tools: Excel, pivot tables, formulas

Framework: Growth Accounting

Health check: Quick Ratio

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Where growth actually comes from

A top-line "users are up" chart hides more than it shows. This analysis decomposes **weekly active users** into new, retained and resurrected users, then uses the **Quick Ratio** to check whether growth is healthy or just masking churn.

Excel | Pivot tables | Formulas | Weekly Active Users | Quick Ratio

Key Insights:

Source data is **device-level:** one record per unique device active in a week.

Duplicate (Device_ID, Week) records were removed first, so **WAU is never inflated.**

The model rests on one identity: **WAU(t) = New + Retained + Resurrected.**

Every bar, decomposed

Each bar is one week of WAU, split into new, retained and resurrected users, so the chart shows what drives growth, not just the trend.



Data and method

Key fields

- Device_ID: unique device
- Week / Week_Num: label and chronological index
- First_Week: first week seen
- Is_New_User: helper flag

Cleaning

Duplicate (Device_ID, Week) records removed before any calculation.

Core identity

WAU(t) = New + Retained + Resurrected, computed per week.

Formulas (1 of 2)

Metric	How it is built	Formula
Weekly Active Users	Pivot table: rows Week, values count of unique Device_ID	Pivot count
New Users	Helper column; filter Is_New_User = 1	=IF(Week_Num = First_Week, 1, 0)
Retained Users	Active this week and not new	Retained = WAU - New

Formulas (2 of 2)

Metric	How it is built	Formula
Resurrected Users	Returning after a gap; negatives floored at zero	MAX(0, WAU - New - Retained)
Churned Users	Active last week, not back this week	MAX(0, WAU(prev) - Retained)
Quick Ratio	Is growth outrunning churn?	(New + Resurrected) / Churned

Reading the Quick Ratio

< 1	~ 1	> 1	~1 to 1.2
Declining	Flat / fragile	Growing	This cohort
			Just inside growing, barely

New and resurrected users are only marginally outpacing churn.

Business insight

Trajectory

Moderate, inconsistent growth: positive overall, not compounding week over week. Early traction, limited stickiness.

Drivers

Mostly new acquisition. Churn stays consistently high; retention is stable but flat, a sign of weak habit formation.

Resurrection

Sporadic: some latent value in the product, but nowhere near enough to offset churn.

Fix retention first

01

Onboarding

Get the first session to a clear first win.

→

02

Early activation

Convert new users before they churn in the first weeks.

→

03

Repeat-usage triggers

Give users a reason to come back, building habit.

Reducing early-stage churn lifts the Quick Ratio directly, letting new-user growth compound instead of leaking out the bottom.

Thank you for reading.

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DISCLAIMER

All figures and definitions are from my growth accounting analysis. The Quick Ratio range is as reported on the portfolio page; the three retention levers are recommendations, not measured results.

Growth Accounting